

Re: Response to Market Competition Subindex Evaluation

Dear Professor Phillips-Sawyer,

Thank you for your thorough and thoughtful evaluation of the Market Competition subindex. Your critiques are valuable, and we appreciate the care you took in reviewing this element of our Index. We have made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

Dynamic Competition and Productive Efficiency

“I would first offer my agreement with the strategy but I would temper that with some hesitancy about the singular use of economists’ tools and thinking. I agree that the challenge for any country is both productive efficiencies and dynamic competition. My reservation seeps in as we consider what we know about dynamic competition and how to bolster or protect it. As I understand it, this is a relatively new field and, at least in antitrust economics, somewhat contentious with regard to its predictive power. We would like to prioritize incentives to invest in innovative technologies, but it can be quite difficult to identify these. Of course, this is not my insight, it is a core concern for the economics profession as covered, in part, by the Schumpeter versus Arrow debate. In turn, the more difficult it becomes to quantify and predict outcomes, the more we rely on our own policy preferences. Or, along a similar line of thinking, there may be more than one rationale for protecting market-oriented competition.”

We agree that the incentives for sustained market competition are difficult to predict and that relying solely on economic forecasting tools can bias results toward implicit policy preferences. To best create an unbiased cross-national metric, our framework measures the capacity of the institutional environment to enable competition rather than specific policy prescriptions through the three components: creative destruction (addressed in detail in the specific creative destruction section), regulatory structures, and human capital competitiveness. While the creative destruction component is an explicit measure of turnover and renewal, the other two are measures of political, legal, and organizational capacity for competition. Our goal is to balance the “economist’s tools” contained in creative destruction with broader institutional metrics.

Creative Destruction

“Two issues came to mind when thinking about this measure. First, in a very large and sophisticated economy like the United States, one might worry that the market share instability might not be driving the economic growth.”

Regarding our use of GDP growth as a proxy for market share instability, we agree that market instability correlates with other unobserved explanatory variables. However, we lean on the logic in the literature to use GDP growth alongside measures of firm entry and exit to best approximate market turnover. As Mazzucato and Tancioni argue, periods of faster growth reflect uneven technological change and shifting relative positions among firms rather than uniform expansion.¹

¹Mariana Mazzucato, Massimiliano Tancioni. Indices that Capture Creative Destruction: Questions and Implications. In: *Revue d’économie industrielle*, vol. 110, 2e trimestre 2005. pp. 199-220.

While this approach abstracts from industry-level dynamics, it follows the insight that economy-wide change can capture the turbulence associated with creative destruction. Paired with firm entry and exit rates, GDP growth functions as part of a broader strategy to proxy creative destruction rather than assert a specific causal channel.

“Second, and more importantly, even where the correlation might be revealing of creative destruction leading to economic growth, many developed countries attempt to mitigate the most painful aspects of creative destruction (bankruptcy and firm closure). But it’s not clear that we should give a higher numerical value for prioritizing market stability at least in the short run.”

We agree that many advanced economies mitigate the immediate disruptions associated with creative destruction and create error in typical measurements of creative destruction. To focus on the measurable inputs to competitive dynamism rather than privileging either stability or disruption, the Creative Destruction component combines firm entry and exit rates, R&D expenditure, and survey-based indicators of competitive intensity. Our PCA methodology facilitates the accurate combination of these datasets regardless of their respective correlative directions. However, we recognize that the relationship between the effects of creative destruction and state policy responses warrants further exploration as new datasets become available. In particular, we are interested in the World Bank’s new B-Ready series, which includes indicators of market competition, but its early roll-out only covers 60 countries of their expected 150+. We will be sure to revisit B-Ready when its full country coverage is complete.

“This brings me to a second point about the creative destruction component of the sub-index. While I agree with the focus on R&D expenditures as a predictor of (or necessary input for) innovation and growth, to what extent does it matter where the investment originates or how it flows through firms? Recently, I wrote a short essay for University of Chicago’s Booth School of Business ProMarket website, ‘Whither Pro Competitive Industrial Policy?’ One thing that I emphasized in that piece is the recent economic evidence regarding federal funds targeted at particular research areas (in contrast to broad tariffs to foster infant industries or growth). And, critically, this is an important policy tool of developed world countries, rather than developing countries. There is some exciting new research in this area and it should be considered for inclusion in the index.”

We agree that targeted public investment in research plays a meaningful role, particularly in advanced economies. Unfortunately, there is not yet a globally comparable dataset that disaggregates public R&D to target state funding. As our goal is to measure Market Competition and, in this case, Creative Destruction, across as close to 176 countries as we can get, datasets that address only a handful of countries introduce significant biases (and often get thrown out by our PCA process). Our current framework captures aggregate R&D expenditure as a share of GDP, university-industry collaboration, and survey-based indicators of government support for long-term R&D investment. As global datasets that disaggregate public R&D arise, we will certainly revisit this question.

The LLM-based approach used by Juhasz, Lane, and Rodrik (2023), mentioned in your ProMarket article, is a significant advancement in how industrial policy can be identified and quantified across countries and sectors. However, incorporating this measure into our Market Competition subindex presents a conceptual challenge: because the GCI normalizes variables to 0-10 by country-year pair, greater industrial policy prevalence would mechanically translate into a higher score. This risks embedding a normative judgment that more industrial policy is inherently despite any underlying variation in effectiveness, design, or objective.

Political Considerations

“First, and as I have written above, I am concerned that the Global Capitalism Index relies too heavily on economic analysis to the detriment of political considerations. While I realize that the choice was both considered and intentional, it nevertheless still seems like an important omission. In considering market competition, for example, we might focus on Silicon Valley as a locus of entrepreneurial activity, investment, and innovation. But, I would argue that we can only partially understand Silicon Valley’s sustained success as an industrial district through the measurements offered in the sub-index. We would not be able to capture the DARPA funds that fostered early developments or the open-source inter-firm collaboration that characterized some of the early computer design and build spaces.”

We acknowledge that certain qualitative features of industrial ecosystems are difficult to capture through quantitative cross-national indicators and are more amenable to case-study analysis. We view this as an inherent limitation of any global index and one that underscores the importance of using the GCI alongside qualitative scholarship. It is worth noting that our Market-Supporting Policy subindex is designed to measure state capacity and willingness to sustain markets, including indicators of government support and funding to start-ups.

Country Tranches

“My suggestion is to consider providing the Index results within various tranches. For example, I wondered if we might learn more from the relative measures by grouping countries according to a variety of schemes, such as income level. Another alternative might be thinking about low-growth versus high-growth countries. In that way, perhaps the Index provides a more nuanced analysis for what a particular country is trying to achieve, relative to where it has been.”

We agree that doing tranced analyses can be a vital part of extracting both correlational and causal lessons from this Market Competition subindex and the GCI as a whole. We have already begun developing analytic processes that tranche countries not only by output variables like income level, growth, V-Dem democracy score, etc, but also through a cluster analysis methodologies that identify tranches of countries within our own data. We anticipate using tranced analyses in all research projects once the baseline GCI scores are complete.

Herfindahl-Hirschman Index

“I have not seen the Hirschman Herfindahl Index used to this effect. It is primarily used to gauge market concentration within a properly defined market. So, I was surprised to see it used in this other way in the Data Outline.”

Upon further review, we agree that the problems created by using the Herfindahl-Hirschman Index for cross-national comparison outweighs its benefits. Therefore, we have removed the from the Market Competition subindex. As you note, the HHI is conventionally applied to measure concentration within a defined product or service market, and our application of it risks introducing confusion. We originally intended to use it as a measure of exposure to broad international competition in markets, but ultimately this risks conflating trade openness with market competition. The

global HHI index is included in our Free Flow of Goods and Capital subindex to capture export diversity and the openness to international trade markets.

Use of Executive Survey Data

“Also, I was surprised that executive surveys would be used to determine whether markets are competitive—executives tend to overestimate the amount of competition that they face.”

We acknowledge the bias inherent in executive surveys. Our reliance on World Economic Forum Executive Opinion Survey data reflects a practical tradeoff between data reliability and geographically comprehensive measurements of less easily observed concepts. To mitigate this bias, we employ several safeguards.

First, our reliability scoring system assigns WEF survey data lower reliability weights (typically 3.3–4.1 out of 5) compared to institutional datasets such as World Governance Indicators or V-Dem measures (which receive scores of 4.5–5.0). These reliability scores directly attenuate the influence of survey-based indicators in the PCA estimation through our correlation adjustment procedure.

Second, we combine survey data with objective indicators wherever possible. If survey responses do not co-vary meaningfully with the other more objective indicators in a subindex, they will typically receive lower factor loadings and contribute less to the final component score.

We understand that the use of executive surveys creates issues, and we aim to design, collect, and incorporate our own surveys in future years. However, we believe that the methodological rigor and geographic coverage of the WEF survey data outweigh its pitfalls. By adjusting for its potential biases using the methods described above, we believe that these datasets contain enough signal to merit their inclusion.

Concluding Remarks

Your substantive critiques are legitimate and well-taken. In particular, the removal of the HHI from this subindex and giving more attention to the relationship between productive efficiency and dynamic competition strengthen this subindex. We would be happy to discuss any of these points further, and we hope to continue this productive dialogue about how to conceptualize and measure market competition.

Sincerely,

UVA Democracy and Capitalism Lab